

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

Assessment Tool Weightage: 10%

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

1. Detection Performance Comparison

Two object detection models produce the following results:

Model	True Positives	False Positives	False Negatives
A (Viola-Jones)	80	20	30
B (YOLO)	90	15	20

(a) Compute precision and recall for both models.

(b) Compare their performance and state which model is better for object detection.

2. Motion Estimation Error Analysis

Two motion estimation techniques produce displacement errors (in pixels):

Frame Method A Method B

1	2	3
2	3	5
3	2	4

(a) Compute average error for both methods.

(b) Compare and evaluate which method is more accurate.

3. Optical Flow Magnitude Comparison

Optical flow magnitudes (in pixels/frame):

Region Method A Method B

R1	5	8
R2	6	10
R3	5	9

- (a) Compute average motion magnitude for both methods.
- (b) Compare and evaluate which method captures stronger motion.

4. Depth Estimation Comparison (Stereo vs SfM)

Estimated depth values (in meters):

Object Stereo Vision Structure from Motion

A	2.0	2.5
B	3.0	3.8
C	4.0	5.0

Actual depths: A=2, B=3, C=4

- (a) Compute absolute error for both methods.
- (b) Compare and evaluate which method is more accurate.

5. Pattern Recognition Accuracy Comparison

Two recognition systems produce results:

System Correct Predictions Total Samples

Traditional	85	100
Deep Learning	92	100

- (a) Compute accuracy for both systems.
- (b) Compare and evaluate which system performs better.

6. Precision-Recall Trade-off Analysis

Two models produce:

Model TP FP FN

A	70	30	20
B	85	25	25

- (a) Compute precision and recall for both models.
- (b) Compare and evaluate which model is more balanced.

7. F1-Score Comparison

Two models:

Model Precision Recall

A	0.8	0.6
B	0.75	0.75

- (a) Compute F1-score for both models.
- (b) Compare and evaluate which model is better.

8. Optical Flow Consistency

Motion values:

Frame	Method A	Method B
1	4	6
2	5	8
3	6	9

(a) Compute average motion.

(b) Compare consistency and evaluate which is more stable.

9. Detection Accuracy Comparison

Results:

Model	Correct	Total
A	180	200
B	170	200

(a) Compute accuracy.

(b) Compare and evaluate.

10. Error Rate Comparison

Results:

Method	Errors	Total
A	20	100
B	10	100

(a) Compute error rate.

(b) Compare and evaluate.

11. Motion Estimation Variance

Errors:

Frame	A	B
1	2	6
2	3	7
3	2	5

(a) Compute mean error.

(b) Compare variability and evaluate stability.

12. IOU Comparison

IOU values:

Object	A	B
1	0.6	0.8
2	0.7	0.85
3	0.65	0.9

(a) Compute average IOU.

(b) Compare and evaluate detection quality.

13. Recall Comparison in Detection

Results:

Model	TP	FN
A	60	40
B	80	20

- (a) Compute recall.
(b) Compare and evaluate sensitivity.

14. Depth Estimation Error (Squared)

Predicted:

Object	A	B	Actual
1	2.5	2.2	2
2	3.5	3.1	3
3	4.5	4.2	4

- (a) Compute squared error.
(b) Compare and evaluate accuracy.

15. Recognition Precision Comparison

Results:

System	TP	FP
A	90	10
B	85	5

- (a) Compute precision.
(b) Compare and evaluate performance.

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation ; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

SIMATS ENGINEERING

Assessment Tool 3: Comparative Analysis — Answer Sheet

Course Name: Computer Vision | Course Code: ITA05

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Name: VIGNESHWAR S V Reg. No: 192321135

3. Optical Flow Magnitude Comparison

Given: Optical flow magnitudes (pixels/frame)

Region	Method A	Method B
R1	5	8
R2	6	10
R3	5	9

(a) Average Motion Magnitude

Method A:

$$\text{Average_A} = (5 + 6 + 5) / 3 = 16 / 3$$

Average_A = 5.33 pixels/frame

Method B:

$$\text{Average_B} = (8 + 10 + 9) / 3 = 27 / 3$$

Average_B = 9 pixels/frame

(b) Comparison and Evaluation

Method	Average Motion
A	5.33 pixels/frame
B	9 pixels/frame (higher)

Since $9 > 5.33$, Method B captures stronger motion.

Conclusion: Method B is better for detecting stronger motion, as its average optical flow magnitude (9 pixels/frame) is higher than Method A's (5.33 pixels/frame).

4. Depth Estimation Comparison — Stereo vs SfM

Given: Estimated depth values (m); Actual depths: A = 2, B = 3, C = 4

Object	Actual	Stereo Vision	SfM
A	2.0 m	2.0 m	2.5 m
B	3.0 m	3.0 m	3.8 m
C	4.0 m	4.0 m	5.0 m

(a) Absolute Error

Formula: Absolute Error = | Predicted – Actual |

Object A – Stereo: $|2.0-2.0| = 0$ SfM: $|2.5-2.0| = 0.5$

Object B – Stereo: $|3.0-3.0| = 0$ SfM: $|3.8-3.0| = 0.8$

Object C – Stereo: $|4.0-4.0| = 0$ SfM: $|5.0-4.0| = 1.0$

Object	Stereo Error	SfM Error
A	0 m	0.5 m
B	0 m	0.8 m
C	0 m	1.0 m

Average Error:

Stereo Vision = $(0+0+0)/3 = 0$ m

SfM = $(0.5+0.8+1.0)/3 = 2.3/3 = 0.77$ m

(b) Comparison and Evaluation

Stereo Vision has an average error of 0 m, while SfM has an average error of approximately 0.77 m. Therefore, Stereo Vision is more accurate.

Conclusion: Stereo Vision provides perfect estimates for the given data, whereas SfM consistently overestimates depth. Hence, Stereo Vision is the more accurate method for this dataset.

6. Precision-Recall Trade-off Analysis

Given:

Model	TP	FP	FN
A	70	30	20
B	85	25	25

Formulas: Precision = $TP / (TP + FP)$ Recall = $TP / (TP + FN)$

Model A

Precision: $P_A = 70 / (70+30) = 70/100 = 0.70 = 70\%$

Recall: $R_A = 70 / (70+20) = 70/90 = 77.78\%$

Model B

Precision: $P_B = 85 / (85+25) = 85/110 = 77.27\%$

Recall: $R_B = 85 / (85+25) = 85/110 = 77.27\%$

Comparison

Model	Precision	Recall
A	70%	77.78%
B	77.27%	77.27%

Model A has slightly higher recall, but its precision is considerably lower. Model B has almost equal precision and recall, both around 77.27%.

Balanced Model (F1-score)

$$F1 = 2PR / (P + R)$$

$$F1_A \approx 73.68\% \quad F1_B \approx 77.27\%$$

Conclusion: Model B is more balanced. It provides a better balance between precision and recall and also has the higher F1-score.

9. Detection Accuracy Comparison

Given:

Model	Correct	Total
A	180	200
B	170	200

(a) Accuracy

$$\text{Formula: Accuracy} = (\text{Correct} / \text{Total}) \times 100$$

$$\text{Accuracy_A} = (180/200) \times 100 = 90\%$$

$$\text{Accuracy_B} = (170/200) \times 100 = 85\%$$

(b) Comparison

Model	Accuracy
A	90%
B	85%

$$\text{Difference: } 90 - 85 = 5\%$$

Conclusion: Model A is better because it achieves 90% accuracy, which is 5 percentage points higher than Model B's 85%. Therefore, for an application where overall detection accuracy is the main criterion, Model A is preferred.

14. Depth Estimation Error — Squared

Given:

Object	Predicted A	Predicted B	Actual
1	2.5	2.2	2
2	3.5	3.1	3
3	4.5	4.2	4

$$\text{Formula: Squared Error} = (\text{Predicted} - \text{Actual})^2$$

Object 1

Method A: $(2.5-2)^2 = (0.5)^2 = 0.25$

Method B: $(2.2-2)^2 = (0.2)^2 = 0.04$

Object 2

Method A: $(3.5-3)^2 = (0.5)^2 = 0.25$

Method B: $(3.1-3)^2 = (0.1)^2 = 0.01$

Object 3

Method A: $(4.5-4)^2 = (0.5)^2 = 0.25$

Method B: $(4.2-4)^2 = (0.2)^2 = 0.04$

Object	Method A	Method B
1	0.25	0.04
2	0.25	0.01
3	0.25	0.04

Mean Squared Error (MSE)

$MSE_A = (0.25+0.25+0.25)/3 = 0.25$

$MSE_B = (0.04+0.01+0.04)/3 = 0.09/3 = 0.03$

Comparison

Method	MSE	Accuracy
A	0.25	Lower
B	0.03	Higher

Lower MSE indicates better depth estimation.

Conclusion: Method B is more accurate. It has an MSE of 0.03, much lower than Method A's 0.25. Therefore, Method B produces depth estimates much closer to the actual values.

Code of Conduct

I, **VIGNESHWAR S V (Reg. No: 192321135)**, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____